

## Ex 1 : Create a Spring Web Project using Maven

```
"C:\Program Files\Eclipse Adoptium\jdk-17.0.12-hotspot\bin\java.exe" ...
Inside main method...
Inside main method...

  _ _ _ _ _
 / \ / \ _ _ _ _ _ ( _ ) _ _ _ _ _ \ \ \ \ \
( ( ) _ _ _ _ _ | ' _ | ' _ | ' _ \ _ _ | \ \ \ \ \
 \ \ _ _ _ _ _ | _ ) | | | | | | | ( _ | ) ) ) )
  ' | _ _ _ _ _ | _ | _ | _ | _ _ _ _ _ | / / / /
=====|_|=====|_ _ _ _ _ / _ _ _ _ _

:: Spring Boot ::                (v3.5.3)

2025-07-12T09:31:25.216+05:30 INFO 10280 --- [spring] [ restartedMain] com.cognizant.spring.Application : Starting Application using Java 17.0.12 with PID 10280 (C:\
2025-07-12T09:31:25.224+05:30 INFO 10280 --- [spring] [ restartedMain] com.cognizant.spring.Application : No active profile set, falling back to 1 default profile: "
2025-07-12T09:31:25.406+05:30 INFO 10280 --- [spring] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add
2025-07-12T09:31:25.407+05:30 INFO 10280 --- [spring] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'lo
```

Output :

## Ex 2: Spring Core – Load Country from Spring Configuration XML

### Application.java

```
package com.cognizant.spring;
```

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```
public class Application {
```

```
    private static final Logger LOGGER = LoggerFactory.getLogger(Application.class);
```

```
    public static void main(String[] args) {
        LOGGER.debug("START");
        displayCountry();
        LOGGER.debug("END");
    }
```

```
    public static void displayCountry() {
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
        Country country = (Country) context.getBean("country", Country.class);
        LOGGER.debug("Country : {}", country.toString());
    }
```

```
}
```

## Country.java

```
package com.cognizant.spring;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class Country {
    private static final Logger LOGGER = LoggerFactory.getLogger(Country.class);

    private String code;
    private String name;

    public Country() {
        LOGGER.debug("Inside Country Constructor.");
    }

    public String getCode() {
        LOGGER.debug("Inside getCode()");
        return code;
    }

    public void setCode(String code) {
        LOGGER.debug("Inside setCode()");
        this.code = code;
    }

    public String getName() {
        LOGGER.debug("Inside getName()");
        return name;
    }

    public void setName(String name) {
        LOGGER.debug("Inside setName()");
        this.name = name;
    }

    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "]";
    }
}
```

## country.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
                           http://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="country" class="com.cognizant.spring.Country">
        <property name="code" value="JP"/>
        <property name="name" value="Japan"/>
    </bean>

</beans>
```

Logback.xml

```
<configuration>
    <appender name="STDOUT" class="ch.qos.logback.core.ConsoleAppender">
        <encoder>
            <pattern>%d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n</pattern>
        </encoder>
    </appender>

    <root level="debug">
        <appender-ref ref="STDOUT"/>
    </root>
</configuration>
```

**Output :**



```
Run Application x
"C:\Program Files\Eclipse Adoptium\jdk-17.0.12.7-hotspot\bin\java.exe" ...
09:55:14.778 [main] DEBUG com.cognizant.spring.Application - START
09:55:14.860 [main] DEBUG o.s.c.s.ClassPathXmlApplicationContext - Refreshing org.springframework.context.support.ClassPat
09:55:15.006 [main] DEBUG o.s.b.f.xml.XmlBeanDefinitionReader - Loaded 1 bean definitions from class path resource [countr
09:55:15.042 [main] DEBUG o.s.b.f.s.DefaultListableBeanFactory - Creating shared instance of singleton bean 'country'
09:55:15.043 [main] DEBUG com.cognizant.spring.Country - Inside Country Constructor.
09:55:15.090 [main] DEBUG com.cognizant.spring.Country - Inside setCode()
09:55:15.090 [main] DEBUG com.cognizant.spring.Country - Inside setName()
09:55:15.102 [main] DEBUG com.cognizant.spring.Application - Country : Country [code=JP, name=Japan]
09:55:15.103 [main] DEBUG com.cognizant.spring.Application - END

Process finished with exit code 0
```

# Hello World RESTful Web Service

## Web.xml

```
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
    http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd"
    version="3.1">

    <display-name>Spring Learn</display-name>

    <servlet>
        <servlet-name>dispatcher</servlet-name>
        <servlet-class>org.springframework.web.servlet.DispatcherServlet</servlet-class>
        <init-param>
            <param-name>contextConfigLocation</param-name>
            <param-value>/WEB-INF/spring-servlet.xml</param-value>
        </init-param>
        <load-on-startup>1</load-on-startup>
    </servlet>

    <servlet-mapping>
        <servlet-name>dispatcher</servlet-name>
        <url-pattern>/</url-pattern>
    </servlet-mapping>

</web-app>
```

## Spring-servlet.xml

```
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:context="http://www.springframework.org/schema/context"
```

```
xmlns:mvc="http://www.springframework.org/schema/mvc"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    http://www.springframework.org/schema/beans/spring-beans.xsd
    http://www.springframework.org/schema/context
    http://www.springframework.org/schema/context/spring-context.xsd
    http://www.springframework.org/schema/mvc
    http://www.springframework.org/schema/mvc/spring-mvc.xsd">
```

```
<context:component-scan base-package="com.friend.spring_auth" />
```

```
<mvc:annotation-driven />
```

```
</beans>
```

### **HelloController.java**

```
package com.friend.spring_auth.controller;
```

```
import org.slf4j.Logger;
```

```
import org.slf4j.LoggerFactory;
```

```
import org.springframework.web.bind.annotation.*;
```

```
@RestController
```

```
public class HelloController {
```

```
    private static final Logger LOGGER = LoggerFactory.getLogger(HelloController.class);
```

```
    @GetMapping("/hello")
```

```
    public String displayHello() {
```

```
        LOGGER.info("BEGIN: displayHello()");
```

```
        String result = "Hello All!";
```

```
        LOGGER.info("FINISH: displayHello()");  
        return result;  
    }  
}
```

**Output :**

Hello All!

## **REST - Country Web Service**

### **spring.xml**

```
<bean id="in" class="com.friend.spring_auth.model.Country">  
    <property name="code" value="IN" />  
    <property name="name" value="India" />  
</bean>
```

### **Country.java**

```
package com.friend.spring_auth.model;
```

```
public class Country {  
    private String code;  
    private String name;  
  
    public Country() {}  
  
    public Country(String code, String name) {  
        this.code = code;  
        this.name = name;  
    }  
}
```

```

        public String getCode() { return code; }

        public void setCode(String code) { this.code = code; }

        public String getName() { return name; }

        public void setName(String name) { this.name = name; }
    }

```

### **CountryController.java**

```

package com.friend.spring_auth.controller;

import com.friend.spring_auth.model.Country;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.*;

@RestController

public class CountryController {

    private static final Logger LOGGER = LoggerFactory.getLogger(CountryController.class);

    @RequestMapping("/country")
    public Country fetchCountryIndia() {
        LOGGER.info("BEGIN: fetchCountryIndia()");
        ApplicationContext context = new ClassPathXmlApplicationContext("spring.xml");
        Country country = (Country) context.getBean("in");
        LOGGER.info("FINISH: fetchCountryIndia()");
        return country;
    }
}

```

```
}
```

**Output :**

```
{
```

```
  "code": "IN",
```

```
  "name": "India"
```

```
}
```

## **REST - Get country based on country code**

### **Country.xml**

```
<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.springframework.org/schema/beans
    http://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean id="countryList" class="java.util.ArrayList">
    <constructor-arg>
      <list>
        <bean class="com.friend.spring_auth.model.Country">
          <property name="code" value="IN"/>
          <property name="name" value="India"/>
        </bean>
        <bean class="com.friend.spring_auth.model.Country">
          <property name="code" value="US"/>
          <property name="name" value="USA"/>
        </bean>
        <bean class="com.friend.spring_auth.model.Country">
          <property name="code" value="UK"/>
          <property name="name" value="United Kingdom"/>
        </bean>
      </list>
    </constructor-arg>
  </bean>
</beans>
```



```
        </constructor-arg>
    </bean>
</beans>
```

### **CountryService.java**

```
package com.friend.spring_auth.service;
```

```
import com.friend.spring_auth.model.Country;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Service;
```

```
import java.util.List;
```

```
@Service
```

```
public class CountryService {
```

```
    public Country fetchCountryByCode(String code) {
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
        List<Country> countries = (List<Country>) context.getBean("countryList");

        return countries.stream()
            .filter(c -> c.getCode().equalsIgnoreCase(code))
            .findFirst()
            .orElse(null);
    }
}
```

### **CountryController.java**

```
package com.friend.spring_auth.controller;
```

```
import com.friend.spring_auth.model.Country;
import com.friend.spring_auth.service.CountryService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;
```

```
@RestController
```

```
public class CountryController {
```

```
    @Autowired
```

```
    private CountryService service;
```

```
    @GetMapping("/countries/{code}")
```

```
    public Country getCountry(@PathVariable String code) {
```

```
        return service.fetchCountryByCode(code);
```

```
    }
```

```
}
```

**Output :**

```
{
```

```
  "code": "IN",
```

```
  "name": "India"
```

```
}
```